

Generation Interconnection System Impact Study Report

For Queue Project Q17-0847

New Hollandale 230 kV

April 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0847. The Interconnection Customer proposes a 678.8 MW natural gas combined cycle facility interconnecting at the New Hollandale 230 kV substation in the Ohio area. The request was received on August 14, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 291 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,949,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
REDFORD - NEW HOLLANDALE 138 kV Line	ALGODONES - BAY PT 345 kV	128.6	241	16.0
CENTRAL - REDFORD 138 kV	BAY PT - ALGODONES SUBSTATION 345 kV	127.2	228	16.0
CENTRAL - BINGEN 138 kV L	BAY POINT - ALGODONES 345 kV line	126.0	217	16.0
NEW HOLLANDALE 100 kV/1 kV XFMR	REDFORD - NEW HOLLANDALE 138 kV	111.4	131	5.4
WAGNER LAKE - NEW HOLLANDALE 230 kV	BAY PT - OAKLAND AVE SUBSTATION 230 kV	111.2	469	50.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
REDFORD - NEW HOLLANDALE 138 kV Line	Rebuild line with higher-capacity conductor	\$1,457,000	97% (\$1,413,000)
REDFORD - CENTRAL 138 kV line	Rebuild line with higher-capacity conductor	\$1,713,000	97% (\$1,661,000)
CENTRAL - BINGEN 138 kV L	Rebuild line with higher-capacity conductor	\$1,225,000	97% (\$1,188,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
NEW HOLLANDALE 100/1 kV Autotransformer	Replace transformer with higher-rated unit	\$3,036,000	71% (\$2,155,000)
WAGNER LAKE - NEW HOLLANDALE 230 kV	Rebuild line with higher-capacity conductor	\$10,317,000	73% (\$7,532,000)
	Total allocated to project		\$13,949,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.